Deft Scheduling of Dynamic Cloud Workflows with Varying Deadlines via Mixture-of-Experts

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Introduction

Dynamic cloud workflow scheduling is crucial for modern cloud services, where workflow tasks must be executed under limited cloud computing resources, cost budgets, and service deadlines. It is also increasingly **relevant to LLM-based agents**, whose planning, tool use, and code execution form workflow-like pipelines that require efficient and reliable resource allocation.

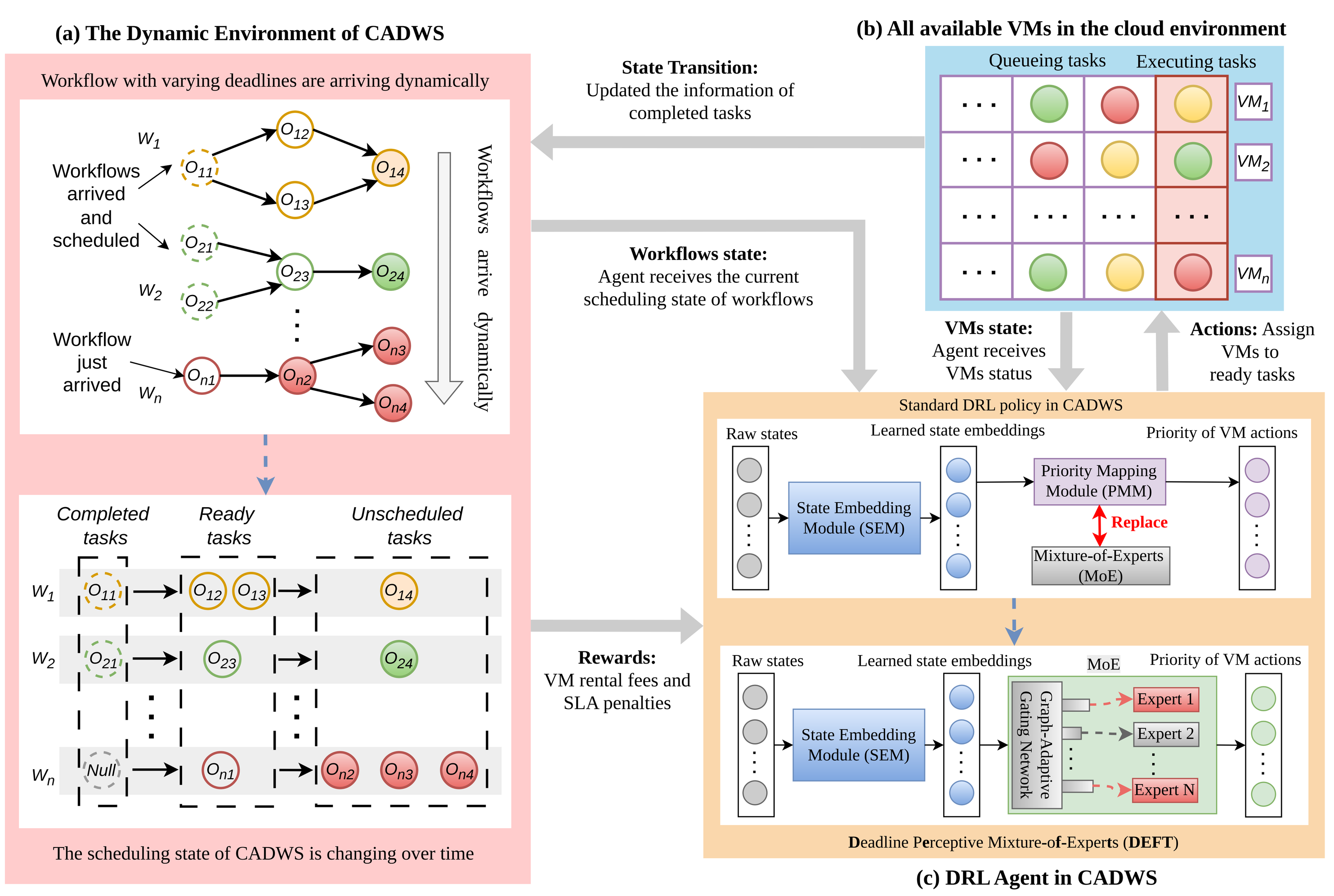


Figure 1. The scheduling process of dynamic cloud workflows via DRL

Challenges:

- Workflows arrive dynamically, while cloud resources change over time.
- Arriving workflows have different levels of deadline/time urgency.
- A single policy is often insufficient for such diverse scheduling scenarios.

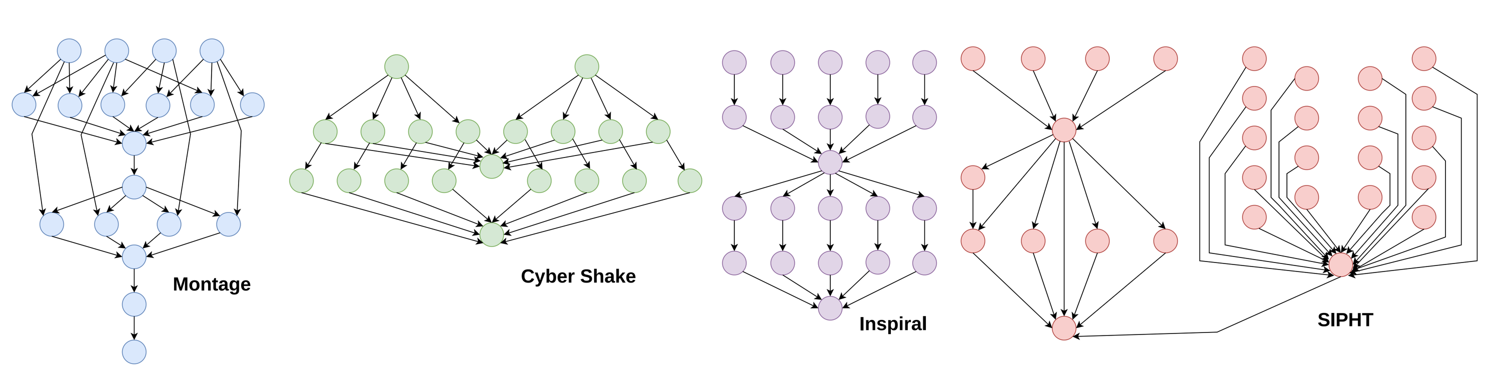


Figure 2. The cloud workflows studied in this work

Contributions

- **First MoE architecture for dynamic cloud workflow scheduling.** DEFT replaces the conventional single-path policy mapping module with multiple deadline-specialized experts.
- **Graph-adaptive gating network.** DEFT uses workflow DAG structure, ready-task features, VM states, and deadline information to perform fine-grained expert selection via cross-attention.
- **Two-phase training strategy.** Experts are first pre-trained on different deadline regimes, then jointly fine-tuned with a gating network for adaptive decision making.

Problem Definition

Setting. Dynamic cloud workflows arrive online and will be scheduled on heterogeneous cloud VMs. Each workflow W_i is modeled as a DAG:

$$W_i = (\mathcal{O}_{W_i}, \mathcal{C}_{W_i}),$$

where $\mathcal{O}_{W_i}$ is the task set and $\mathcal{C}_{W_i}$ denotes precedence constraints. A task becomes ready once all its predecessors are completed. Each workflow arrives at time a_i and has an SLA deadline:

$$d_i = a_i + \gamma \cdot \mathrm{minMakespan}(W_i), \tag{1}$$

where $\gamma \geq 1$ controls deadline tightness.

Resources. The VM resources in the cloud are heterogeneous and changing over time:

$$\mathcal{M} = \{m_1, \dots, m_{|\mathcal{M}|}\}, \tag{2}$$

where different VM m_j has different processing speed v_j and hourly cost c_j.

MDP. We formulate CADWS as an MDP $(\mathcal{S}, \mathcal{A}, \Pr, R)$ with return:

$$R(\tau) = -\sum_{t=0}^{T-1} C_t^{\mathrm{vm}} - C_T^{\mathrm{sla}}(\mathcal{W}), \tag{3}$$

where C_t^{vm} is the VM rental cost and $C_T^{\mathrm{sla}}(\mathcal{W})$ is the SLA penalty.

Objective. Learn an optimal policy

$$\pi^* = \arg\max_{\pi} \mathbb{E}_{\tau\sim\pi}[R(\tau)], \tag{4}$$

Methodology

DEFT Framework. DEFT replaces a monolithic policy with multiple deadline-specialized experts. A **graph-adaptive gating network** selects the most suitable expert based on DAG structure, ready-task features, VM states, and deadline urgency. **Cross-attention** is used to compute expert scores and dynamically activate the expert that best matches the current scheduling context.

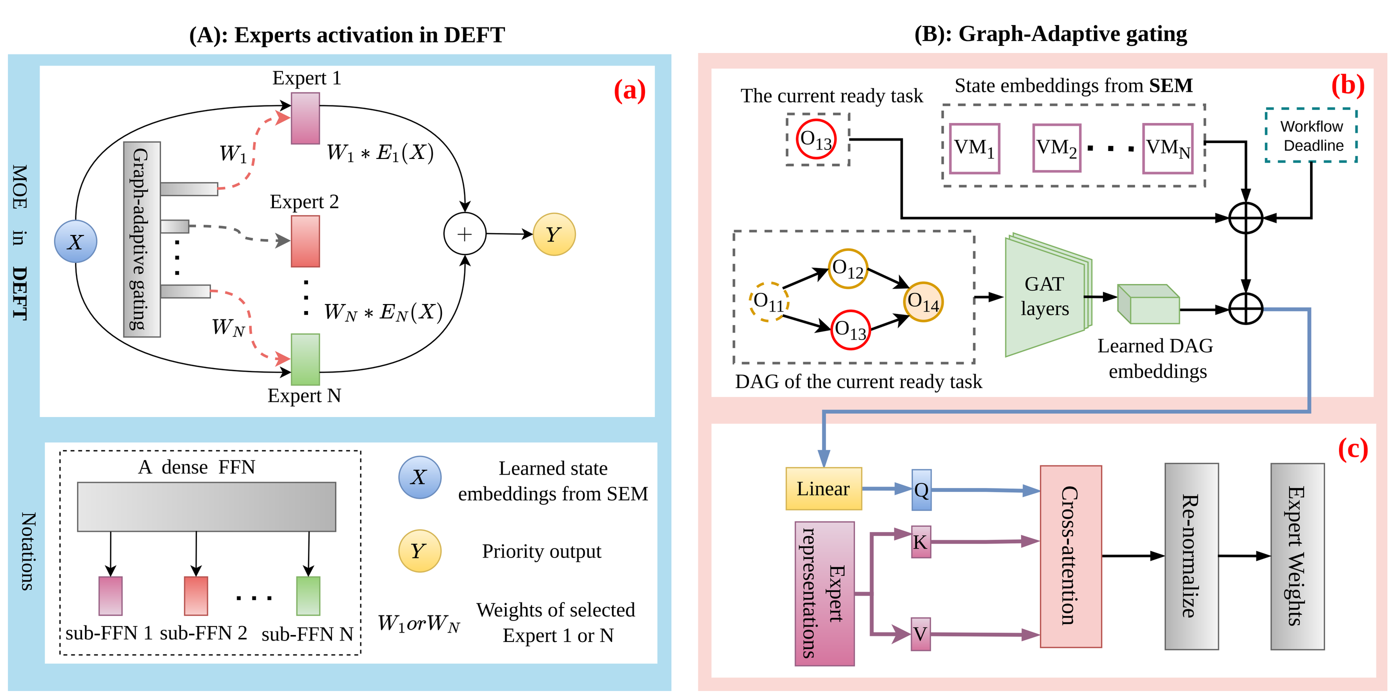


Figure 3. The MoE and graph-adaptive gating network in DEFT

Two-Phase Training. (1) Pre-train experts under fixed deadline regimes to learn specialized behaviors. (2) Jointly fine-tune experts, gating, and state embedding modules on mixed-deadline instances for adaptive end-to-end scheduling.

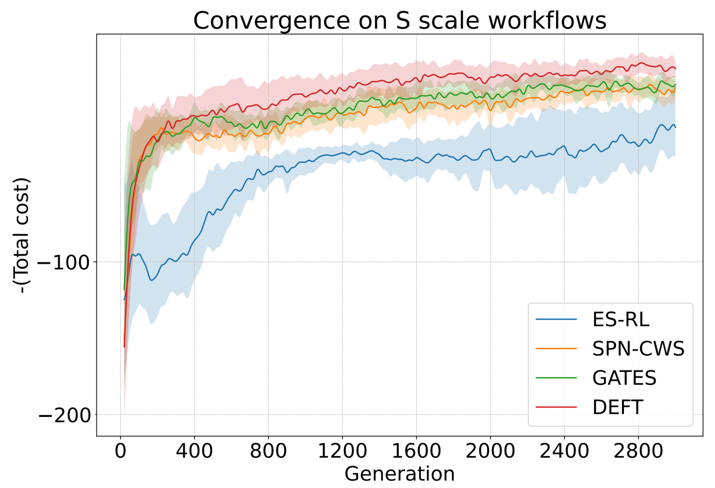
Experiments

Main Results (Table 1). DEFT delivers the lowest total cost across S/M/L settings. Its advantage over other baselines increases with workflow scale, indicating better generalization.

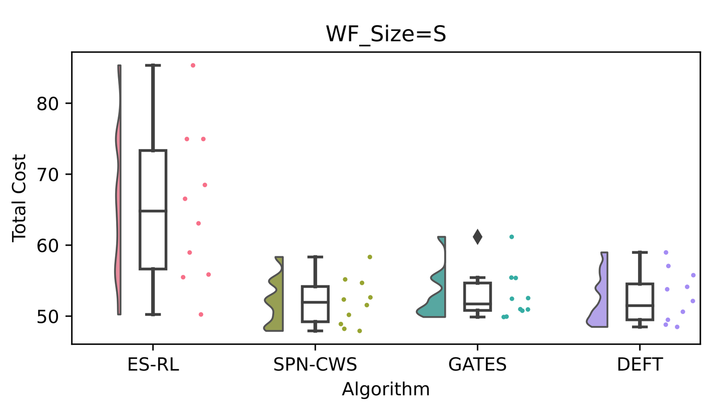
Table 1. Total cost (mean) with VM fees / SLA penalties under dynamic deadlines.

Scenario	ProLis		GRP-HEFT		ES-RL		SPN-CWS		GATES		DEFT	
	Cost	VM/SLA	Cost	VM/SLA	Cost	VM/SLA	Cost	VM/SLA	Cost	VM/SLA	Cost	VM/SLA
$\langle S\rangle$	183.74	79.01 / 104.73	297.58	297.58 / **0.00**	65.39	29.12 / 36.27	54.99	23.31 / 31.68	52.95	**20.65** / 32.30	**52.46**	21.01 / 31.45
$\langle M\rangle$	304.03	176.34 / 127.69	495.64	495.64 / **0.00**	109.23	63.75 / 45.48	87.69	44.05 / 43.64	97.76	55.34 / 42.42	**86.60**	**41.06** / 45.54
$\langle L\rangle$	755.21	279.43 / 475.78	1064.34	1064.34 / **0.00**	225.46	170.45 / 55.01	149.26	90.64 / 58.62	195.65	145.20 / 50.45	**137.69**	**70.88** / 66.81

Convergence and Stability. DEFT converges faster and reaches a lower final cost than all baselines, showing improved learning efficiency and stability.



(a) Convergence among DRL baselines



(b) Stability among DRL baselines

Why Top-1 in MoE ?

Table 2. Top-k blending can select a VM that no expert actually prefers (expert weights w_1=0.4, w_2=0.6).

	Action probability distributions			Who prefers this VM?		
VM	Expert 1 $\pi^{(1)}(a)$	Expert 2 $\pi^{(2)}(a)$	Mixed $\pi_{\mathrm{mix}}(a)$	Expert 1	Expert 2	Top-k ($k=2$) mixing
A	0.10	0.05	0.07	–	–	–
B	**0.45**	0.10	0.24	Yes	–	–
C	0.05	**0.50**	0.32	–	Yes	–
D	0.40	0.35	**0.37**	–	–	Yes

Conclusion: CADWS requires one VM decision per step, mixing multiple experts can blur the action signal. Top-1 preserves expert specialization by selecting the single best-matched expert.

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